2-STOREY RESIDENTIAL BUILDING: FULL CAD PACKAGE

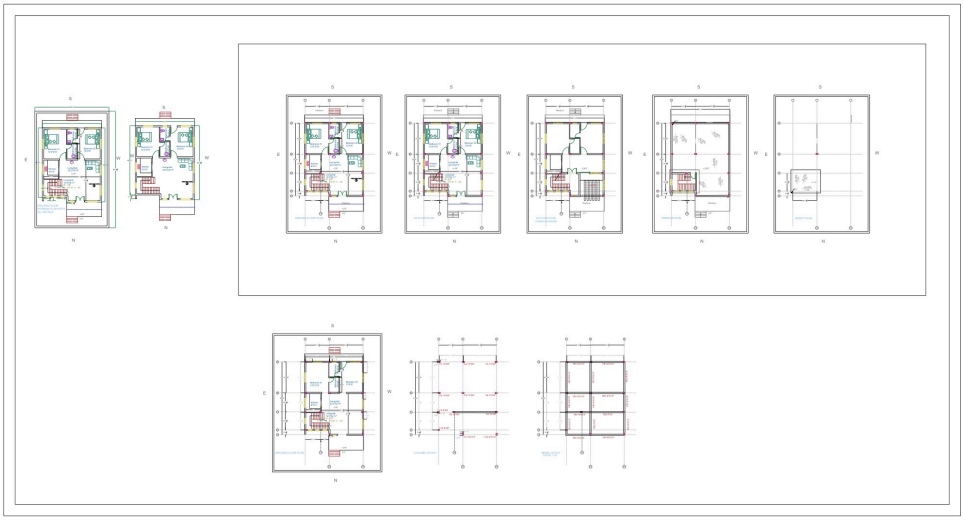
AUTOCAD DRAFTER / DESIGN ENGINEER: RISHABH DEY

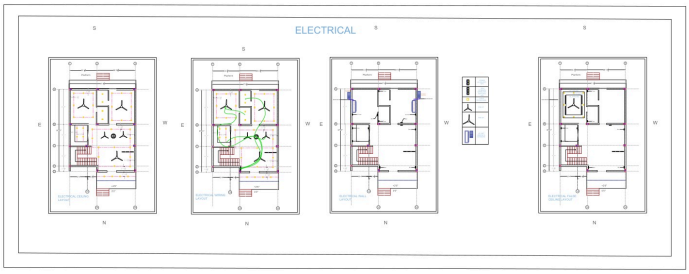
Architectural planning • Structural layout • Electrical coordination

I designed this compact two-storey residence as one complete drawing package, taking it from architectural planning through to structural and electrical coordination.

The set covers a ground-floor working plan and a matching upper floor, each with two bedrooms, two bathrooms, a kitchen, a living/hall zone and a stair, followed by terrace and mumty plans at levels up to +33'6". The structural sheet gives a twelve-column layout (C1 to C12) and a beam layout at level +14', and the electrical series covers ceiling, wiring, wall and false-ceiling layouts with fans, light points, switchboards, power points and AC units. Every view is drawn on one shared grid, so architecture, structure and services can be read against each other.

This dossier records what is visible on the drawings and flags the items to confirm before construction or statutory submission.





Project title used in this document is descriptive; no client name, site address, architect, or issue date was supplied in the source sheets.

1. Project Overview

The supplied set represents a residential building design with a ground-floor working plan, repeated upper-floor planning, terrace and mumty information, a column layout, a beam layout, and electrical ceiling, wiring, wall and false-ceiling layouts. The plans show bedrooms, bathrooms, kitchens, living/hall areas, stairs and external platforms/entry areas.

Project intent

The plan is organized around a compact residential core: sleeping rooms and bathrooms occupy the upper portion of the floor plate, kitchen and living spaces sit alongside the stair and entrance zone, and vertical circulation connects the levels. Repetition of the main room arrangement on upper floors suggests a typical-floor strategy.

Drawing set at a glance

Drawing / sheet	Visible content	Use in this dossier
Sheet 1	Ground-floor working plan with dimensions and room labels	Room planning and dimensional review
Sheet 2	Ground, upper-floor, terrace and mumty plans	Floor-to-floor organization and roof planning
Sheet 3	Ground-floor plan, column layout and beam layout	Structural coordination review
Sheet 4	Electrical ceiling, wiring, wall and false-ceiling layouts	Electrical planning review
Sheet 5	Composite drawing sheet containing architectural, structural and electrical views	Reference index / overview

Basis and assumptions

• All dimensions and room names in the narrative are transcribed from the visible labels; some small fractions and characters are difficult to read in the source PDFs.

• The drawings show levels including 0'0", +2'6", +14'6", +22'6", +26'6", +33'6" and a mumty/terrace arrangement. Their exact datum interpretation should be checked against the original CAD file.

• The drawing set does not provide a site address, north-based site plan, occupancy calculation, material specifications, design loads, or signed engineering calculations.

2. Architectural Planning

2.1 Room arrangement

The principal plan identifies two bedrooms, a kitchen, bathroom spaces, a large living/hall zone, a smaller living/hall or lobby zone, and a stair. Openings are shown along the external walls, with door swings indicating access between rooms and circulation areas.

2.2 Room dimensions shown

Space	Dimension labelled on plan	Approx. area*
Bedroom 01	12'3" × 16'	196 sq ft
Bedroom 02	13' × 16'	208 sq ft
Kitchen	8' × 10'	80 sq ft
Living / hall (main)	22'2" × 21'9"	482 sq ft
Living / hall (secondary)	12'4" × 8'11" (label appears as 8;11")	110 sq ft
Bathrooms	Labels include 5'3" × 7'7"/7'8", 5'3" × 8', and 5'3" × 8' (verify exact allocation)	Varies

*Areas are simple length × width calculations from the displayed nominal dimensions and are not certified carpet/built-up areas. Wall thicknesses, shafts, niches and measurement conventions are excluded.

2.3 Access and circulation

• The stair is placed near the entrance/living zone and provides the principal vertical circulation route.

• Bedroom doors open from the central internal circulation zone; bathroom doors are grouped close to the bedrooms.

• The plan indicates a main entrance with double-leaf door symbols and additional windows/openings along external walls.

• Check that door swings, stair landings and furniture clearances remain unobstructed in the detailed working drawing.

3. Planning Performance & Design Review

Daylight and ventilation

Window symbols appear on several external walls, including bedroom and living areas. This can support daylight and natural ventilation, but actual performance depends on orientation, opening size, adjacent buildings and local climate. The north arrow appears on the plan sheets; the site orientation should be coordinated with a site plan before assessing solar exposure.

Privacy and zoning

The separation of bedrooms from the main living/hall zone provides a basic public/private zoning strategy. Bathroom grouping can simplify plumbing routes, although the exact vertical alignment of soil, waste and vent stacks should be verified across all floors.

Stair information

A stair annotation reads approximately “R - 7"/7.5", T - 10"” and a width annotation appears as “W - 3'6"”. These are recorded as drawing notes only. Confirm the intended riser, tread, clear width, headroom, handrails, landings and applicable code requirements against the actual stair geometry.

Items to resolve during design development

• Confirm whether the upper-floor plans are intended as first and second floors; one view is labelled “1st floor plan” more than once.

• Reconcile the overall external dimensions and the internal dimension strings against the editable CAD drawing before quantity take-off.

• Confirm room labels and bathroom dimensions where text overlaps or is small in the supplied raster/PDF sheet.

• Provide a door/window schedule with type, width, height, sill level and material.

• Add furniture layouts, kitchen counter details, bathroom fixture details and accessibility clearances if required by the project brief.

4. Structural Drawing Review

The structural sheet includes a column layout and a beam layout. Column marks visible on the drawing include C1 through C12, with different size labels; beam labels include FB1 and FB2. The grid references use numbered axes and lettered axes, allowing the structural elements to be coordinated with the architectural plan.

Structural elements identified on sheet

Mark / notation	Visible label	Documentation note
C1–C3	1'3" × 9" (as printed)	Confirm whether this is a column size or a rotated dimension; notation appears atypical.
C4–C6	1'3" × 9" (as printed)	Verify against structural engineer's schedule.
C7, C10	9" × 9" (as printed)	Check exact location and section designation.
C8–C9, C11–C12	Some labels read 1'3" × 9"	Confirm exact sizes from original CAD.
FB1	9" × 1'3"	Beam mark / section label shown repeatedly.
FB2	9" × 2'	Beam mark / section label shown in selected locations.

Structural coordination requirements

• Confirm grid spacing, column dimensions and beam sizes against signed structural calculations and the original CAD file.

• Check that columns do not conflict with doors, windows, kitchen counters, bathroom fixtures or stair geometry.

• Provide foundation layout, footing schedule, reinforcement details, slab thickness, beam-column joint details and material grades; these are not included in the supplied sheets.

• Confirm load paths, seismic/wind design criteria, soil bearing capacity and applicable local structural standards with a qualified structural engineer.

This review is descriptive, not a structural design certification. The visible sizes should not be used for construction without verification.

5. Electrical & Ceiling Layout Review

The electrical sheet contains four views: electrical ceiling layout, electrical wiring layout, electrical wall layout and electrical false-ceiling layout. Symbols and labels indicate fans, lighting points, switches, power points and air-conditioning indoor/outdoor components. The drawing also includes ceiling-level notes and a reference to SB-2.

Elements visible in the drawing

• Ceiling fan symbols are shown in room and living areas; labels include fan sizes of 14" and 30" in selected locations.

• Lighting points and a wiring route are indicated on the electrical wiring plan.

• Switchboards are marked “SB-2” in several positions.

• Power points are shown at selected walls and at air-conditioning locations.

• Indoor and outdoor AC components are identified, with a note referring to AC power.

• Ceiling-related notes include +9' false ceiling, +10' false ceiling and +11' ceiling in the legend area; confirm which note applies to each room.

Information needed for an executable electrical design

• Electrical load schedule, circuit numbering, cable sizes, conduit sizes, breaker ratings and distribution-board location.

• Earthing, bonding, residual-current protection, surge protection and isolation details.

• Dedicated circuits and isolators for air conditioners and other high-load equipment.

• Exact mounting heights for switches, sockets, control points and AC units.

• Coordination between lighting, fans, false-ceiling framing, beams and access panels.

• Legend with complete symbol definitions and a clear separation of power, lighting, data and low-voltage systems.

Do not infer wire sizes or protective-device ratings from the layout alone; they must be designed from actual loads, installation conditions and applicable electrical standards.

6. Construction Coordination & Deliverables

Recommended pre-construction checks

• Confirm drawing scale, units, datum levels and north orientation.

• Verify all overall dimensions and room dimensions against the editable source file.

• Coordinate architectural, structural and electrical drawings on the same grid and revision.

• Prepare plumbing, drainage, water supply and rainwater-pipe drawings; the architectural/terrace sheet shows an RWP notation but not a complete plumbing design.

• Add foundation, slab, reinforcement, stair and lintel details.

• Prepare door/window schedules and finish specifications.

• Confirm waterproofing details for bathrooms, terrace and roof drainage.

• Check local development-control rules, setbacks, fire/life-safety provisions and approval requirements.

• Create a bill of quantities only after dimensions, specifications and drawing revisions are confirmed.

Drawing coordination register

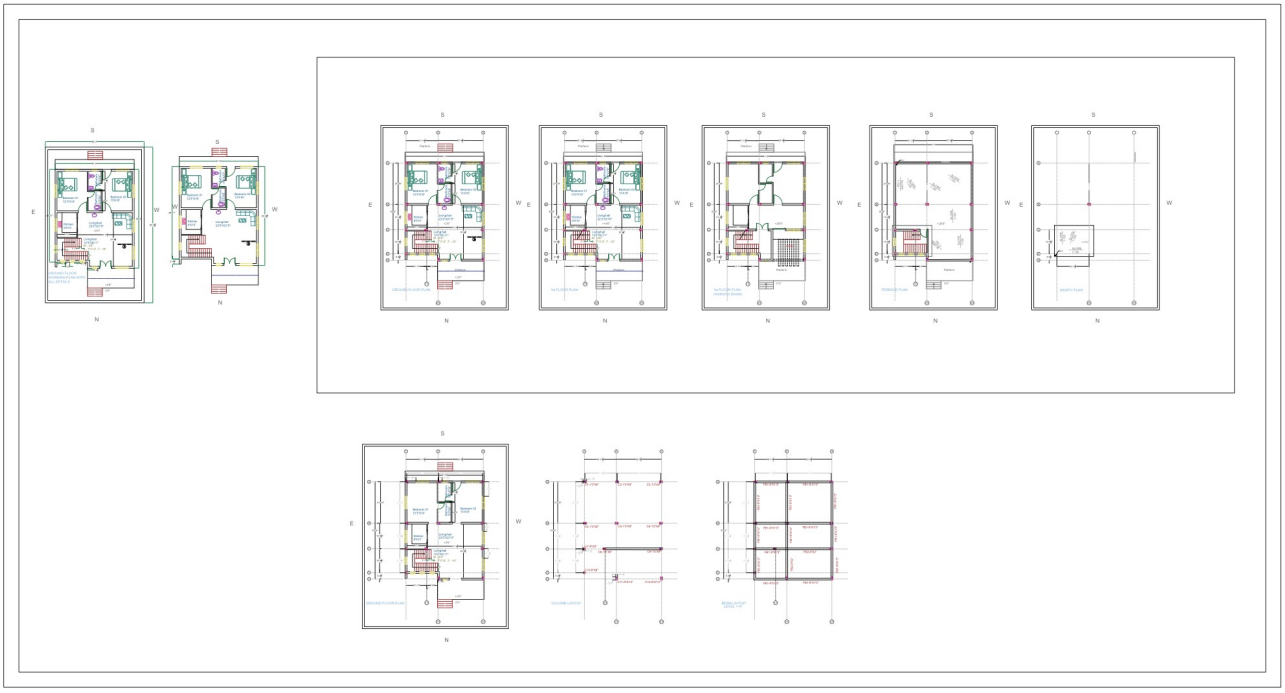
Discipline	Supplied?	Next required output
Architecture	Partial / plan views supplied	Dimensioned plans, elevations, sections, schedules and finishes
Structure	Column and beam layouts supplied	Design calculations, foundations, reinforcement and sections
Electrical	Layout views supplied	Load/circuit schedules, DB details, earthing and specifications
Plumbing / drainage	Not provided as a complete layout	Water supply, soil/waste/vent, rainwater and drainage plans
Site / approvals	Not provided	Site plan, setbacks, area statement and statutory documentation

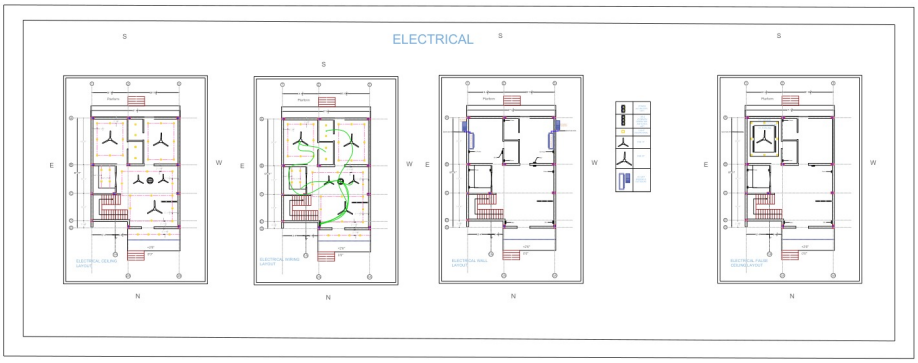
Limitations

This document is a narrative documentation of the supplied drawing sheets, not a replacement for signed architectural/engineering drawings. Some source labels are small or ambiguous in the provided PDF images. No missing specifications, calculations or code compliance claims have been invented. All construction dimensions and technical details must be checked against the original editable drawings and reviewed by the responsible professionals.

Appendix A · Full Plan

Composite source sheet consolidates architectural plans and structural/electrical drawings for cross-reference.





Sheet contents

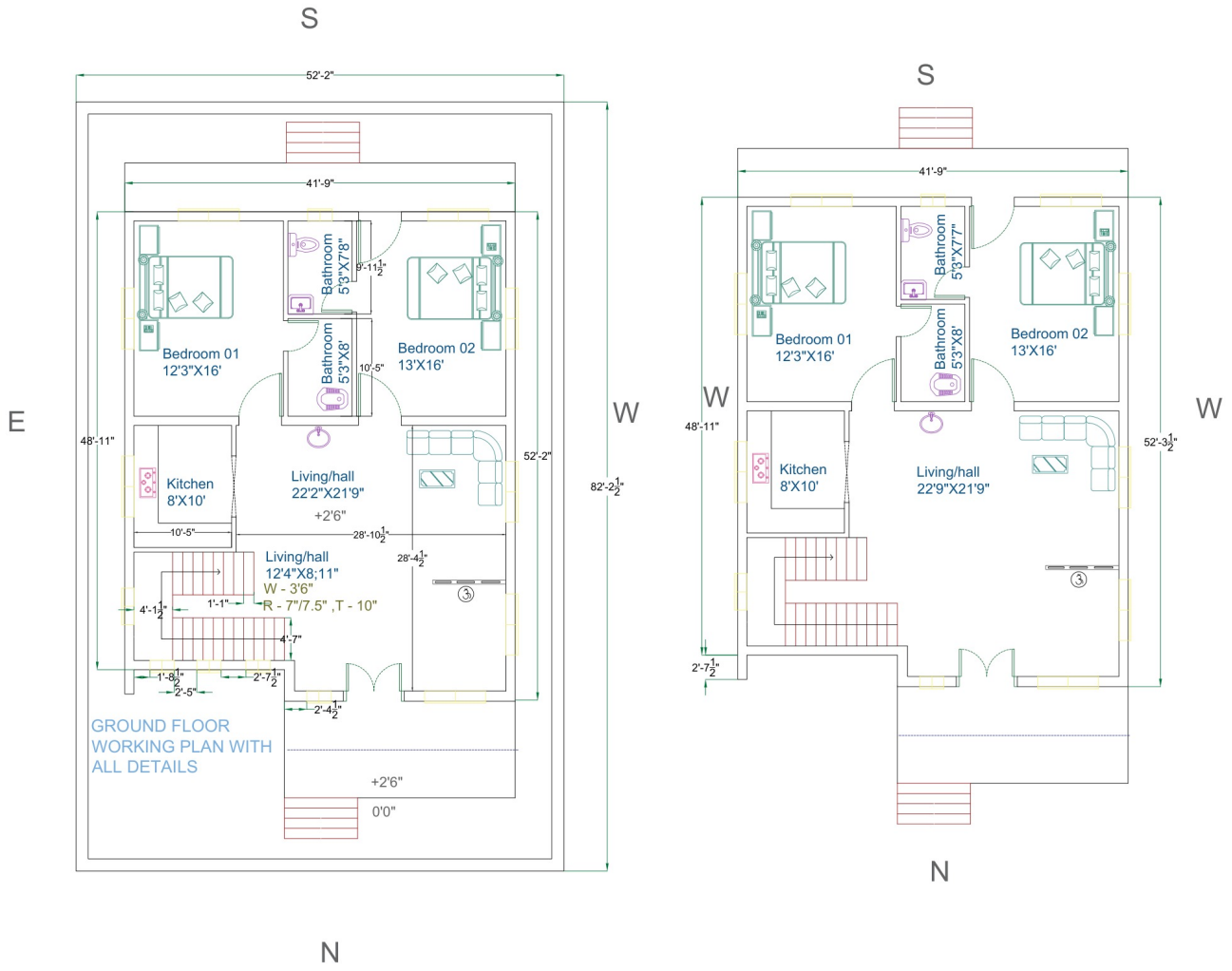
Group	Views	Main information shown
Top-left pair	Ground-floor working plan; upper-floor plan	Room names, dimensions, stair note, entry platform and level marks
Architectural series (middle row)	Ground Floor Plan; 1st Floor Plan; 1st Floor Plan (Additional Details); Terrace Plan; Mumty Plan	Levels 0'0", +2'6", +14'6", +26'6" and +33'6"; slope notes of 1:100 on terrace and mumty
Structural row (lower)	Ground floor plan with grid; Column Layout; Beam Layout (Level +14')	Columns C1 to C12 and beams FB1 and FB2 on grids 1 to 3 and A to D
Electrical sheet	Ceiling, Wiring, Wall and False Ceiling layouts, plus a symbol legend	Fans, light points, switchboards, power points and AC units

Common features

- The sheets hold 14 drawings plus one legend: ten architectural and structural views on the main sheet and four electrical layouts on the electrical sheet.
- Orientation labels read S at the top, N at the bottom, E at the left and W at the right, so the entrance platform faces N.
- All views share one grid: axes 1, 2, 3 (with 1A, 2A, 3A along the entrance side) and A, B, C, D.
- Dimension strings on the plans read 41'-9" (width), 48'-11" and 52'-2" (depth, left and right) and an overall outer figure of 82'-2½".
- The upper floors repeat the ground-floor room arrangement, which supports a typical-floor reading.

Appendix B · Ground-Floor Working Plan

Source sheet includes a ground-floor working plan with room names, dimensions, stair information and entry/platform levels.



Spaces and features read from the ground-floor plan

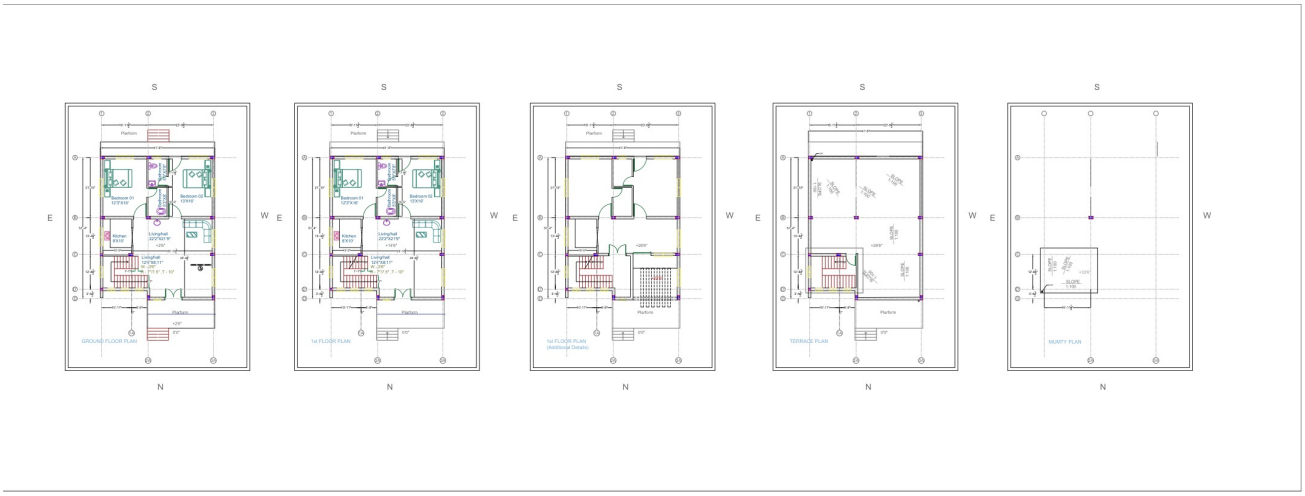
Space	Label on plan	Nominal area	Notes from the drawing
Bedroom 01	12'3" × 16'	196 sq ft	Double bed; windows on two external walls; door opens to the central hall
Bedroom 02	13' × 16'	208 sq ft	Double bed; window on the external wall; door opens to the central hall
Bathroom (upper)	5'3" × 7'8"	40 sq ft	WC and wash basin drawn; sits between the two bedrooms
Bathroom (lower)	5'3" × 8'	42 sq ft	WC drawn; a separate wash basin symbol sits in the hall outside
Kitchen	8' × 10'	80 sq ft	Hob symbol; adjoining dimension 10'-5"
Living / hall (main)	22'2" × 21'9"	482 sq ft	L-shaped sofa and TV unit; level +2'6"; dimension 28'-10½"
Living / hall (secondary)	12'4" × 8'11"	110 sq ft	Holds the stair (W 3'6", R 7"/7.5", T 10"); label prints "8;11" for 8'11"
Entry	Double-leaf door	n/a	Platform at +2'6", ground at 0'0"; red step symbols at both outer ends

Upper-floor differences and items to verify

- The upper floor repeats the layout, but the main hall prints 22'9" × 21'9" (ground: 22'2"), the upper bathroom 5'3" × 7'7" (ground: 7'8") and the depth 52'-3½" (ground: 52'-2"). Confirm which set is correct.
- The stair is drawn as two flights upstairs, against a single run on the ground floor.
- The lower-right room on the ground floor has no name, only a circled “3” marker; confirm its use.

Appendix C · Floor, Terrace & Mumty Plans

Source sheet groups ground-floor, upper-floor, terrace and mumty plans. The repeated upper-floor label should be verified.



Drawings on this sheet

Drawing	Level shown	Visible content
Ground Floor Plan	0'0" at the platform; +2'6" inside	Two bedrooms, two bathrooms, kitchen, living/hall, stair and entry platform; furniture and fixtures drawn
1st Floor Plan	+14'6"	Same room arrangement as the ground floor, with the stair continuing up and step symbols at the platform
1st Floor Plan (Additional Details)	+26'6" marked near the lobby	Walls only, no furniture; stair and a hatched slab area beside the entry lobby; door symbols shown
Terrace Plan	+26'6"	Open roof slab with slope notes of 1:100 in several directions; stair enclosure at the grid C to D zone with a door
Mumty Plan	+33'6"	Small enclosure over the stair at grids C to D; slope 1:100; vertical dimensions 12'-8½" and 3'-4½"

Grid and dimensions

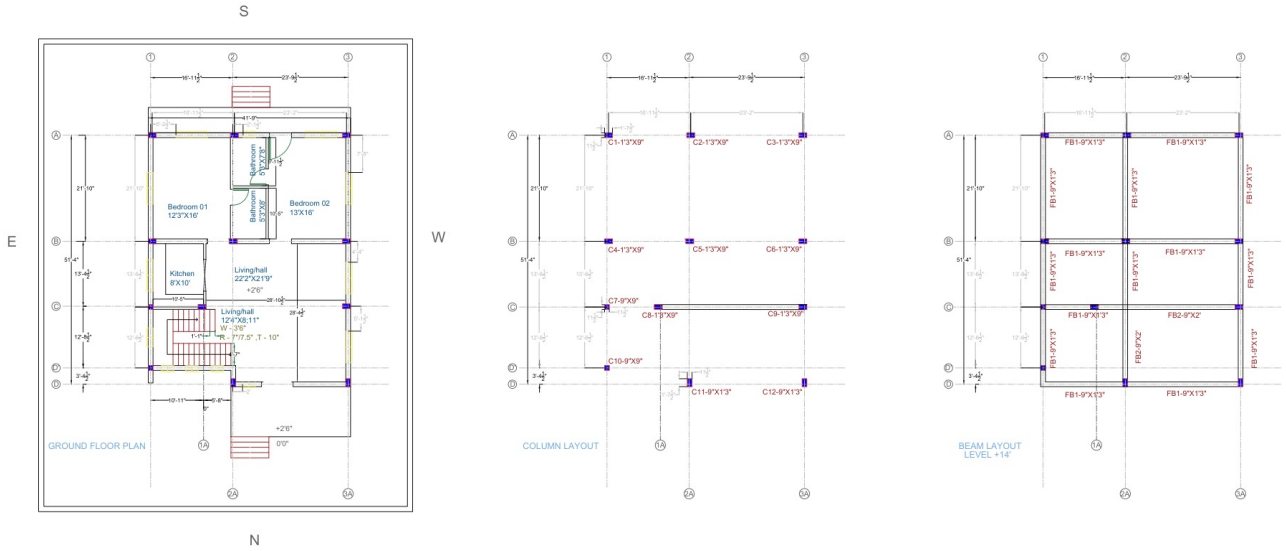
- Axes 1, 2, 3 run across the plan and A, B, C, D run along it; 1A, 2A and 3A mark the entrance side.
- Horizontal bays read 16'-11½" (grids 1 to 2) and 23'-9½" (grids 2 to 3). The overall vertical string reads 51'-4", with 21'-10", 12'-8½" and 3'-4½" legible among the bays.
- Level marks across the set: 0'0", +2'6", +14'6", +26'6" and +33'6". The terrace sits at +26'6" and the mumty roof at +33'6".
- The entry platform label is spelled “Plarform” on the sheet.

To verify

- Two views are titled “1st Floor Plan”. The second carries the +26'6" mark, which matches the terrace level, so confirm what it represents.
- Small bay dimensions are hard to read in the source; check them against the CAD file.
- Drainage routes, parapet heights and waterproofing are not shown on the terrace or mumty plans.

Appendix D · Column & Beam Layout

Source sheet includes the architectural ground-floor plan, column positions and beam layout with grid references and element marks.



Columns as printed on the Column Layout

Mark	Size as printed	Approximate position on the grid
C1, C2, C3	1'3" × 9"	Grid line A, at axes 1, 2 and 3
C4, C5, C6	1'3" × 9"	Grid line B, at axes 1, 2 and 3
C7	9" × 9"	Grid line C, at axis 1
C8	1'3" × 9"	Grid line C, between axes 1 and 2; a continuous line runs from here to C9
C9	1'3" × 9"	Grid line C, at axis 3
C10	9" × 9"	Near grid line D, at axis 1
C11	9" × 1'3"	Grid line D, near axis 2
C12	9" × 1'3"	Grid line D, at axis 3

Beams on the Beam Layout (Level +14')

Mark	Size as printed	Where it appears
FB1	9" × 1'3"	Most beams: along grid lines A, B and C, on axes 1, 2 and 3, and along the D line
FB2	9" × 2'	In the C to D bay: along grid C on the right-hand side and on axis 2 between C and D

Grid and layout

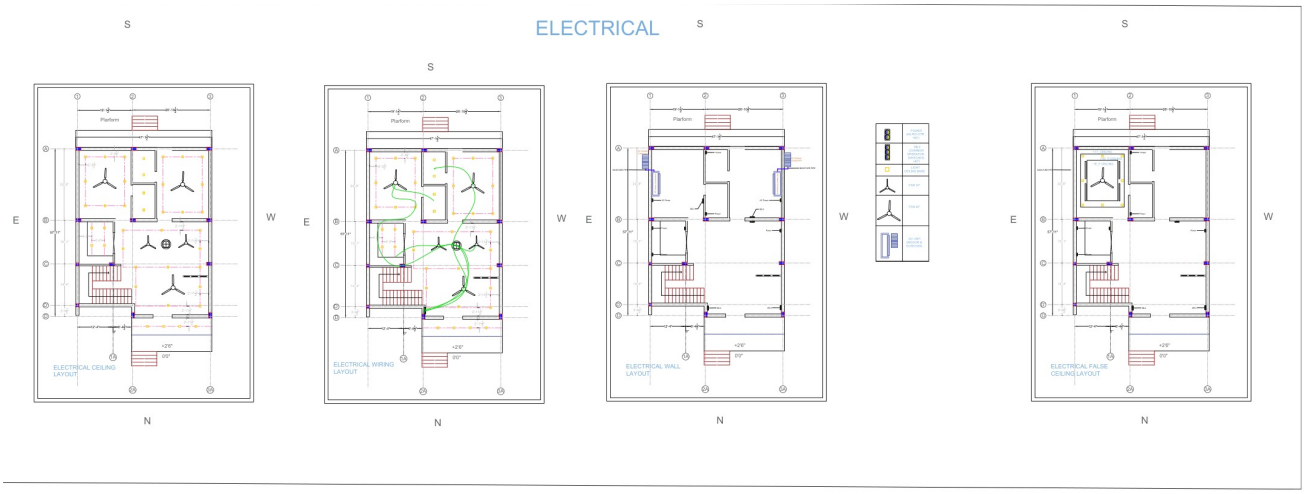
- Spans read 16'-11½" (axes 1 to 2) and 23'-9½" (axes 2 to 3); vertical bays include 21'-10", 12'-8½" and 3'-4½", with a 51'-4" overall string.
- The ground-floor plan on the left of the sheet shows the same walls, rooms and stair against the column positions, so column and wall positions can be compared directly.
- Columns appear as filled squares at the grid intersections. The beam layout forms a closed frame around three bays in each direction.

To verify

- The long line between C8 and C9 along grid C may be a wall or a deep element; confirm its type and section.
- Column marks and sizes should be reconciled with the structural engineer's schedule, since small differences such as 9" × 1'3" and 1'3" × 9" depend on orientation.
- Only a +14' beam layout is shown; roof-level beams, foundations, footings and reinforcement are not part of this set.

Appendix E · Electrical Layouts

Source sheet contains electrical ceiling, wiring, wall and false-ceiling layouts with fans, lighting, switches, power and AC notations.



The four layouts

Layout	What is drawn
Electrical Ceiling Layout	Ceiling fans in both bedrooms and in the living/hall areas, a round fixture in the main hall, and ceiling light points as small squares; dashed outlines mark ceiling bands. About five fans are drawn (visual count).
Electrical Wiring Layout	The same fans and light points, with green wiring routes sweeping from the hall and stair zone out to the bedrooms and entry.
Electrical Wall Layout	Wall-mounted items only: power points, AC power points, SB-2 switchboards, indoor AC units on both bedroom side walls, outdoor AC components outside them, with moisture-pipe lines labelled.
Electrical False Ceiling Layout	A nested rectangular false-ceiling profile with a fan in the Bedroom 01 area, plus the stair, wall and door outlines.

Legend as read

Symbol	Meaning as printed (small text)
Two-gang socket block	Power points (AC, RO and similar), mounting height about +84"
Three-gang switch block	SB-2, common operator switches, mounting height about +40"
Yellow square	Light, ceiling base
Small three-blade fan	Fan 14"
Large three-blade fan	Fan 30"
Blue unit with louvres	AC unit (indoor and outdoor)

Other points visible

- The platform and entry steps are repeated on every layout so switch and point positions can be read against the same grid.
- The ceiling-level notes (+9', +10' and +11') mentioned in Section 5 appear in the false-ceiling and legend areas.

To verify

- The legend text and the mounting heights are small and low resolution; confirm them against the original CAD file.
- No circuit numbers, load schedule, distribution board, earthing or cable and breaker ratings are shown.
- Check fan sizes (14" and 30") and light-point counts room by room before issuing for installation.